

Evaluation theory simulation experiment

Simulation experiment setup

We are considering a case where models are projecting final epidemic size (in an assumed SIR model) for two vaccination coverage scenarios, 30% vs. 50% final coverage. We assume that each model makes n independent projections (e.g., across locations), which differ in their R_0 values and their realized vaccination coverage. Together, we can use these values to get an estimate of final epidemic size.

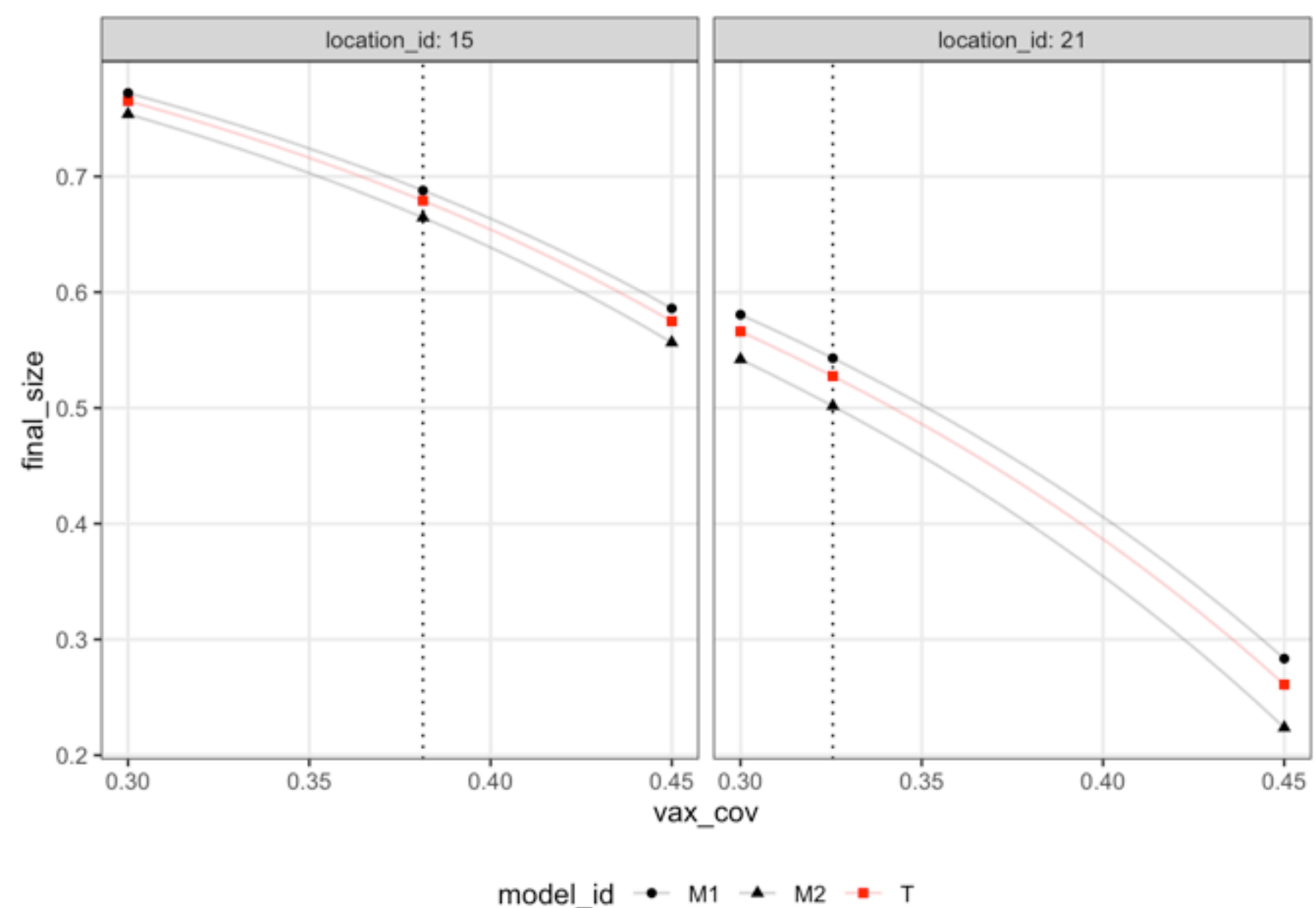
Following this framework, the simulation experiment proceeds as follows:

- define scenario vaccination coverage values, $x_1 = 0.3$ and $x_2 = 0.5$
- define "true" R_0 value and realized vaccination coverage value for each projected location l , $R_{0,l}^* \sim \text{Unif}(2, 3.5)$ and $x_l^* \sim \text{Unif}(0.3, 0.5)$ (so note, in this case, realized vaccination coverage will always fall within the scenario-specified vaccination coverage)
- define some model-specific bias in R_0 , which for the simplest case, we assume is constant across all locations, $b^m \sim \text{N}(0, 0.05)$; then a model's R_0 for a given location is $R_{0,l}^m = R_{0,l}^* + b^m$
- using the `finalsize` package in R, calculate
 - the two values that we typically have when evaluating a set of scenario projections
 - projections: final epidemic size, projected by each model for x_1 and x_2 based on $R_{0,l}^m$, call this $P^m(y|x_i)$ and
 - observations: final epidemic size observed for *realized* vaccination coverage, x^* , using $R_{0,l}^*$, call this $P^*(y|x^*)$
 - the other values we will need when evaluating scenario projections (and testing whether our method works) - see overleaf file for explanation (open circles in Figure 1)
 - projections of final epidemic size from each model for the *realized* scenario, i.e., final epidemic size under vaccination coverage x^* assuming $R_{0,l}^m$, call this $P^m(y|x^*)$
 - true final epidemic size in each of the scenarios that were modeled, i.e., final epidemic size under vaccination coverage x_1 and x_2 assuming $R_{0,l}^*$, call this $P^*(y|x_i)$

Note: this is the simplest simulation setup. Eventually we will modify/complicate the setup to test the sensitivity of the method to various assumptions. For now, let's look at this simple case.

Let's say we have 2 models and 50 locations. The below figure shows what our simulation generates for two random locations: (1) projections from each model for the scenarios of interest, and the realized scenario value (projections in black points, realized scenario value in dotted vertical line), (2) "true" final epidemic size foreach of these scenario values. Note, since this is a simulation, we can also calculate final size at every other scenario value, which is shown with the light lines.

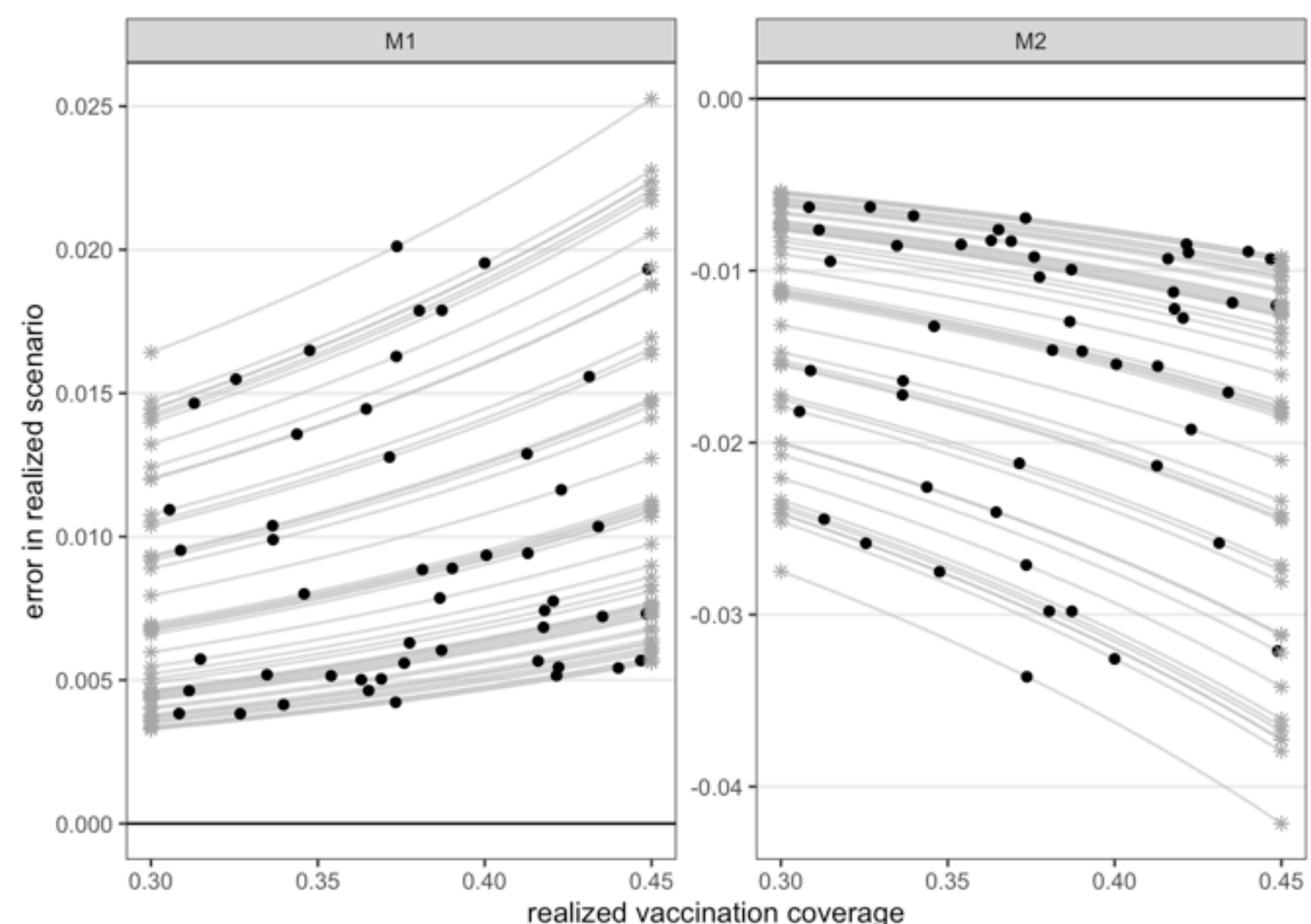
[1] 1



Then, for evaluation, we can calculate the two errors we are interested in:

- error at the realized scenario, $P^m(y|x^*) - P^*(y|x^*)$
- error at the modeled scenarios, $P^m(y|x_i) - P^*(y|x_i)$

In our proposed method (see Overleaf file), we use variation in realized scenario value across locations to estimate model-level error. This plot illustrates the idea. For each model (panel), we plot the realized scenario error from each location as a function of the realized vaccination coverage (black points). These points are drawn from an error function (that we only know/can calculate because this is a simulation), which is $e^m(x) = P^m(y|x) - P^*(y|x)$ (gray lines). Then, the error in the scenarios that we originally modeled is $e^m(x_1)$ and $e^m(x_2)$ (gray stars) - the distribution of these values is what we are ultimately interested in estimating.



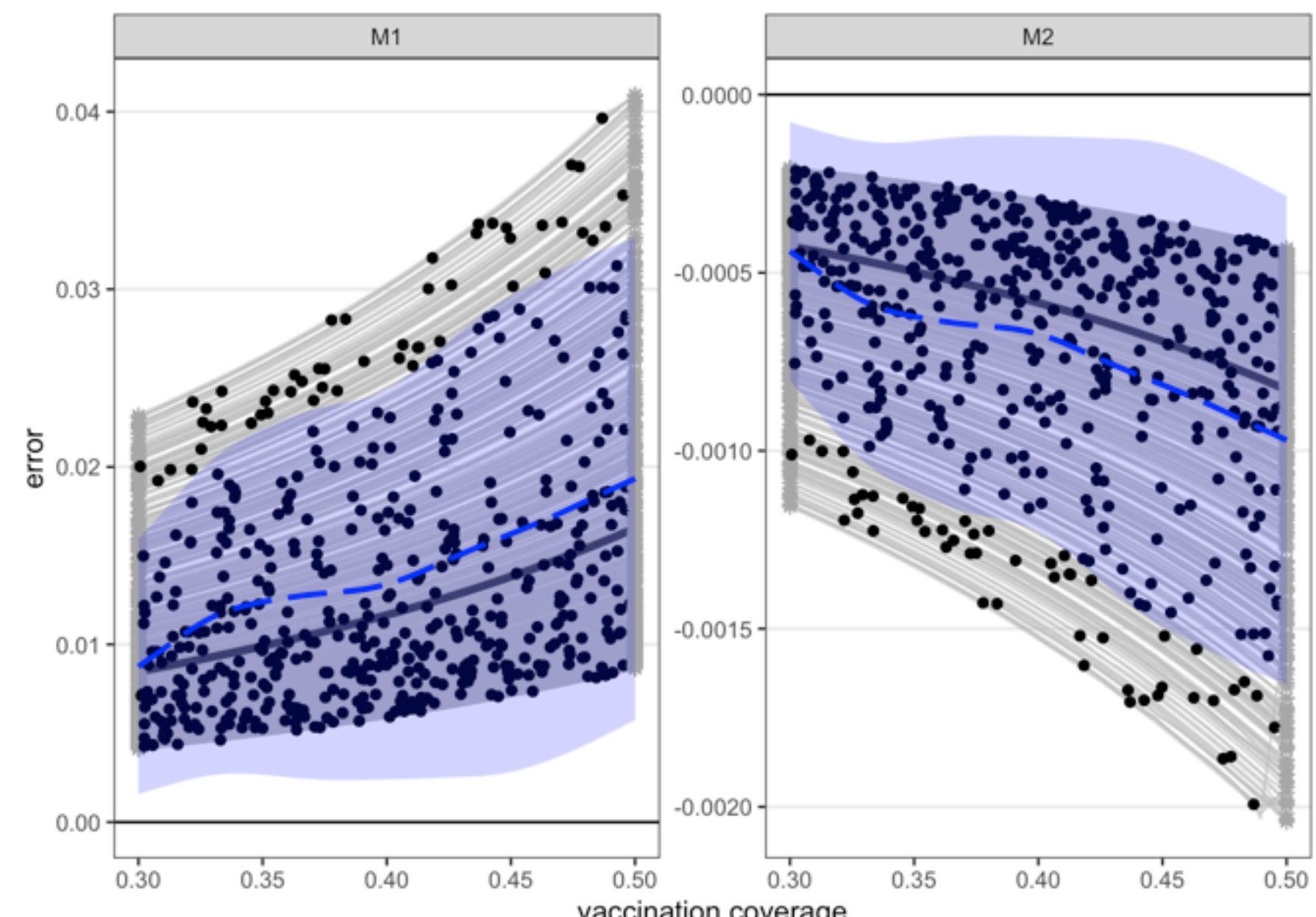
Fitting the error distribution

To estimate these distributions of error at the scenarios that were modeled (i.e., distribution of gray stars at $x_1 = 0.3$ and $x_2 = 0.5$), we propose to fit the error distribution function across the scenario axis using the error at the realized scenario we have across locations. Here, we let $e^m(x) = \beta_0 + g(x) + \epsilon_i(x)$ where $\epsilon_i(x) \sim \text{N}(0, \sigma(x))$. We use a GAM to fit both $g(x)$ and $\sigma(x)$. In particular, I am using the `gamlss` function to do the fitting. Here is the code:

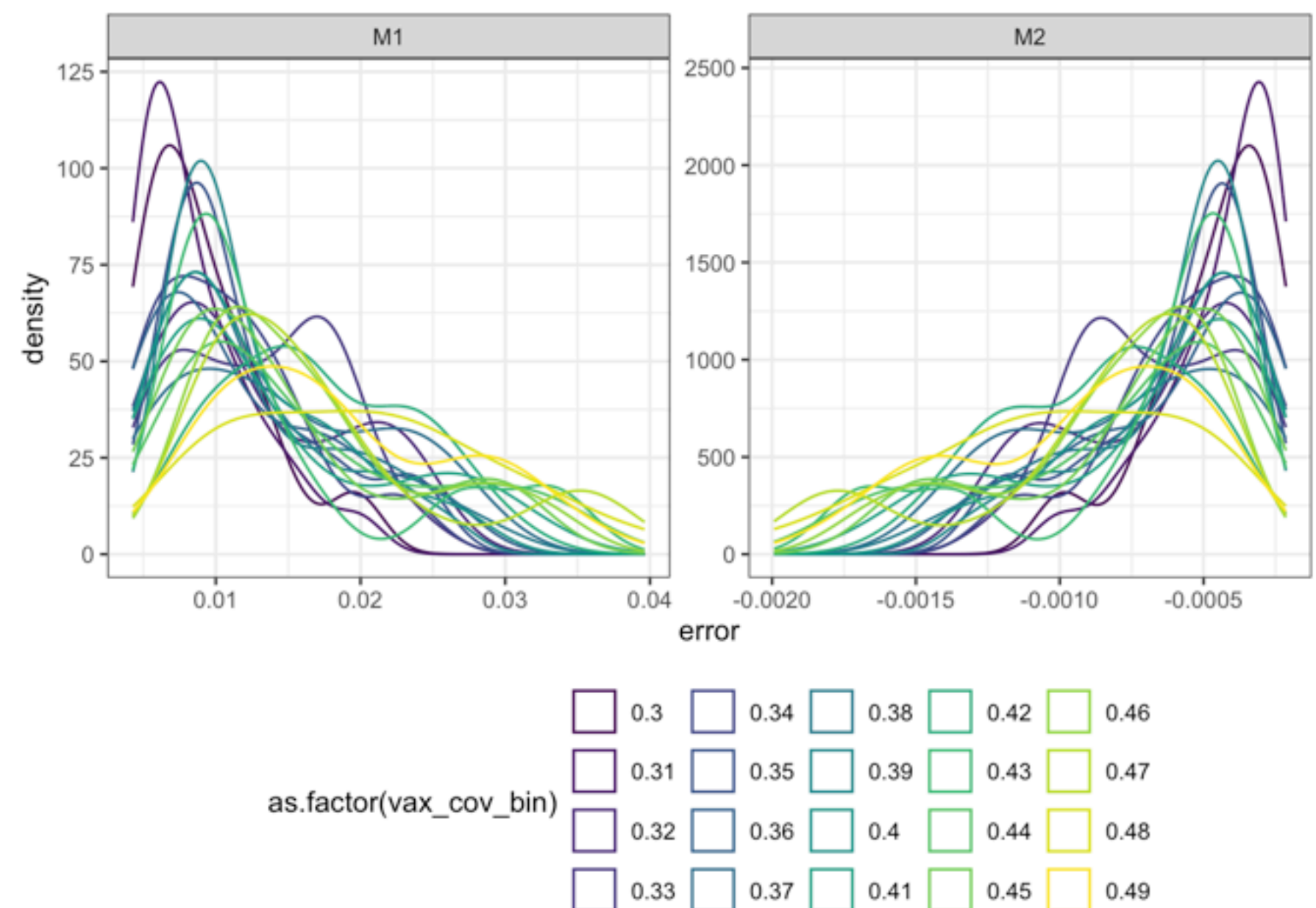
```
# filter to one model, and only error at realized scenario
dat_filt = errors_df %>%
  filter(model_id == paste0("M", i), scenario_id == "T")

# fit GAM with cubic spline for mu and sigma
gam_fit <- gamlss(
  error ~ cs(vax_cov), sigma.formula = ~ cs(vax_cov),
  data = dat_filt, family = N0, trace = FALSE
)
```

I will re-run the simulation with an unrealistic amount of locations ($n = 500$) to illustrate the challenges with fitting this. The blue line and shaded region shows the median and 95% prediction interval of the fitted model.



The fit is bad because our normal distribution assumption fails. Here, I've binned the points based on their realized vaccination coverage (x-axis value). But, a gamma distribution, for example, won't handle the negative errors. And a skew normal doesn't seem to capture the shape either.



Two questions:

- suggestions on how to get improve the fit of this particular example
- suggestions on a general fitting framework (in practice we won't know the gray dots we are aiming for)

I'll also send the above sample data in case you'd like to play around. A quick guide to the simulation data:

```
head(t_large$errors)
```

	model_id	location_id	scenario_id	vax_cov	R0	final_size	true_final_size
1	M1	1	S1	0.3	3.051092	0.8301936	0.8234264
2	M2	1	S1	0.3	3.006275	0.8230870	0.8234264
3	M1	2	S1	0.3	2.096499	0.5610488	0.5397253
4	M2	2	S1	0.3	2.051682	0.5386481	0.5397253
5	M1	3	S1	0.3	2.644476	0.7509524	0.7403127
6	M2	3	S1	0.3	2.599659	0.7397778	0.7403127
			error				relerror
1			0.0067672036				0.0082183465
2			-0.0003394037				-0.0004121847
3			0.0213234795				0.0395080231
4			-0.0010771935				-0.0019958181
5			0.0106396207				0.0143717922
6			-0.0005349468				-0.0007225957

- `model_id`: unique identifier for different models (here M1 or M2 for each model)
- `location_id`: unique identifier for different locations (here integer from 1 to 500)
- `scenario_id`: denotes vaccination scenario (includes S1 and S2 for the scenarios that were modeled, T for the scenario value that was realized in that location, and E for the 'extra' values that allow us to define the lines between the points)
- `vax_cov`: vaccination coverage associated with this scenario
- `R0`: R0 for that location/model
- `final_size`: final epidemic size, given `vax_cov` and `R0`
- `true_final_size`: observed final epidemic size in that location for that vaccination coverage, given the true R0
- `error`: difference between `final_size` and `true_final_size`
- `relerror`: relative difference between `final_size` and `true_final_size`